

DP 302P: Design Practicum-II

Course Code: DP 302P

Course Name: Design Practicum-II

L-T-P-C: 0-0-4-2

Prerequisites: IC 202P: Design Practicum

Intended for: UG

Mutual Exclusion: NA

Distribution: The course will be counted as DE for the students

Approval: 61st BoA (to be reported to Senate)

Preamble

In this course, the teams are asked to improve further the proof-of-concept developed in the design practicum course (IC 202P). The prototype can be used as the basis for developing the final solution. The goal when developing such prototypes is to capture the functions and appearance of the finished product and move towards developing a minimum viable product (MVP). These prototypes are used for testing and evaluation and provide useful information for the user to rank the products or the features. In addition, the prototype developed will formulate a base for a potential startup.

The faculty who have mentored the projects in the earlier DP course (IC 202P) will be allowed to float their projects. The students may continue their earlier DP project or may switch to another project on the recommendation of the project faculty mentor.

Course Modules

The course does not have any dedicated lecture hours. This is a laboratory course; the students will develop the prototype in the labs.

Text Books

1. *Rapid Prototyping: Principles and Applications* by Chee Kai Chua, Kah Fai Leong, Chu Sing Lim. World Scientific Publishing Company Pvt. Ltd.
2. *Engineering Drawing Practice for Schools & College*. SP46:2003.
3. *Illustrating sourcebook of mechanical components* by Robert O. Parmely, P.E., McGraw-Hill.

References

NA

Evaluation Details

- Project Proposal (10 Marks): Students will have to submit a plan of action and the final outcome of the project.
- Midterm Report (20 Marks): Report highlighting the work done till the mid sem.



- Final Report (30 Marks): Reports highlighting the work details, CAD, Codes, and full details of the product.
- Final Presentation (40 Marks): Students will have to present their work as a startup pitch for the developed product. The final presentation will be evaluated by a group of faculty members working on the relevant areas.

Project Selection Criteria & Funds

- Only the top-performing project will be allowed to continue in DP-II.
- These projects will be selected based on their performance on the Open House Day.
- The project will be continued only after approval from its faculty mentor.
- Students will be allowed to switch to another projects subjected to the faculty mentor's approval and the maximum allowable team limit. Preference will be given to students who want to continue the same projects.
- Each year, the project-wise budget will be decided with the approval of the Dean of Academics. The course coordinator will seek approval prior to the start of the semester.